Harkunwar Singh

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Summary

- Honours Computer Science Co-op graduate from McMaster University (April 2026) with a 4.0 GPA in final year, available immediately for full-time Software Engineering, Data, or AI roles.
- Developer with experience building production-style mobile, backend, and ML systems using Python, Java, JavaScript, TypeScript, React Native, FastAPI, PostgreSQL, and cloud platforms.
- Completed a year-long government co-op at the Ministry of Transportation Ontario, contributing to QA validation, process automation, and technical documentation for a live production system.
- Applied ML and NLP experience through CNN-based medical image classification, ensemble modeling, RAG pipelines, LLM integration, and real-world data processing systems.

Education

McMaster University Hamilton, ON

Honours B.Sc., Computer Science (Co-op) *Sept. 2021 - Apr. 2026*

- **Award:** McMaster Faculty of Engineering Award of Excellence Scholarship for Academic Excellence ($3,000).
- **GPA:** Achieved a 4.0 out of 4.0 in the final year of the Honours Computer Science program.
- **Relevant Coursework:** Data Structures and Algorithms, Software Development Life Cycle, Operating Systems, Information Security, Data Analytics, Computer Architecture, Web Programming, Statistics, Discrete Mathematics.

Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, Assembly, HTML, CSS
Frameworks and Libraries: React Native, Expo, Node.js, FastAPI, Streamlit, LangChain, NumPy, Pandas, Scikit-learn, PyTorch, SQLAlchemy
Databases and Cloud: PostgreSQL, MySQL, MongoDB, ChromaDB, Supabase, AWS S3, Microsoft Azure
Tools and Platforms: Git, Docker, Linux, Azure DevOps, Wireshark, Figma
Technical Areas: Machine Learning, Deep Learning, NLP, RAG, LLM Integration, REST APIs, QA Testing, Data Structures and Algorithms, Network Security

Professional Experience

Engineering Associate Co-op North York, ON

Ministry of Transportation Ontario *Aug. 2024 - Aug. 2025*

- Performed end-to-end QA testing and functional validation for a live government Laboratory Information Management System (LIMS) across 5+ release cycles, identifying defects and edge cases before production deployment.
- Managed the full bug lifecycle in Azure DevOps, creating, tracking, and validating tickets through resolution, and coordinating defect fixes with an external vendor development team.
- Developed a Python automation script using pdfplumber and pandas to extract lab result tables from PDFs and convert them to CSV files for LIMS ingestion, eliminating hours of manual data entry across 3+ government lab departments.
- Coordinated QA and testing efforts across 3+ government departments, aligning functional requirements and communicating defect status to the vendor team throughout multiple release cycles.
- Authored bug ticket templates, testing guides, and reporting procedures used for team standardization and onboarding across the QA process.

Projects

NexDerm `[GitHub]` | *React Native, TypeScript, FastAPI, AWS S3, PostgreSQL, PyTorch, SQLAlchemy* Sept. 2025 - Apr. 2026

- Co-developed a cross-platform AI skin lesion screening application for iOS and Android using React Native, Expo, and TypeScript, supporting image capture, real-time predictions, doctor review flows, and PDF report generation.
- Engineered a FastAPI backend deployed on AWS with REST endpoints for authentication, image upload, ML inference, report generation, and doctor review, using SQLAlchemy for database ORM and Supabase PostgreSQL for data persistence.
- Integrated AWS S3 for scalable image storage with IAM-secured bucket policies and Resend for transactional email delivery, supporting a complete end-to-end data and communication flow.
- Trained DenseNet and ResNet CNN models using PyTorch on a 16-class Kaggle skin lesion dataset, combining predictions through an ensemble approach to achieve approximately 70% overall accuracy, with top classes reaching 85-90%.
- Implemented a pytest-based backend test suite with coverage reporting and Jest-based frontend tests, with external services stubbed for deterministic test execution.

Quill `[GitHub]` | *Python, LangChain, Ollama, LLaMA 3.2, ChromaDB, Streamlit* Mar. 2026 - May 2026

- Designed and implemented a fully offline RAG application that lets users chat with any PDF using local LLMs, with no API keys or external data transmission, ensuring complete data privacy.
- Created a document ingestion pipeline using RecursiveCharacterTextSplitter with 1000-token chunks and 200-token overlap, embedded with nomic-embed-text via Ollama, and stored in ChromaDB for persistent local retrieval.
- Applied MMR retrieval to fetch the top-5 most relevant and diverse chunks from 20 candidates, reducing redundancy and improving answer quality across multi-section documents.
- Integrated LLaMA 3.2 via Ollama to generate grounded answers with page-level citations, and evaluated pipeline quality using RAGAS metrics including faithfulness and answer relevancy.

AI Resume-Job Matching System `[GitHub]` | *Python, TF-IDF, Scikit-learn, NLP, Pandas* Jan. 2026 - Present

- Created an explainable NLP-based resume matching system that compares candidate resumes against software engineering job descriptions using TF-IDF vectorization and cosine similarity.
- Designed a preprocessing pipeline with tokenization, stop-word removal, n-grams, and skill extraction using 100+ real-world job postings to improve keyword coverage and reduce noisy matches.
- Developed a scoring module that returns match percentages, matched skills, missing keywords, and prioritized resume improvement suggestions for non-technical users.